Implementation of Shrinkage Estimators for Cosmological Precision Matrices

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Context

Two very different problems share the same statistical bottleneck. In **weather forecasting**, ensemble Kalman filters combine model predictions with observations to keep forecasts on track. In **cosmology**, parameter inference from galaxy surveys turns measurements into estimates of the universe's contents. Both rely on inverting a *covariance matrix* estimated from a small number of simulations or model runs. When the matrix is built from too few samples, the inverse is unstable: tiny noise in the inputs produces huge errors in the outputs. The result is poor forecasts and unreliable cosmological parameters.

Shrinkage estimators address this by blending the noisy estimate with a structured *target*. The trick is choosing the right blend at every step — and choosing it well matters. In numerical weather prediction, an unstable covariance translates into degraded forecasts; in cosmological surveys, it translates into wider error bars on fundamental parameters such as the dark-energy equation of state.

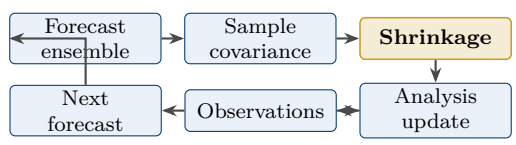
This work integrates a rigorous shrinkage framework into the *sequential* setting of the Ensemble Kalman Filter, where the estimate is recomputed at every assimilation cycle, and asks a practical question: *which estimator should a practitioner pick, and on what basis?*

Objectives

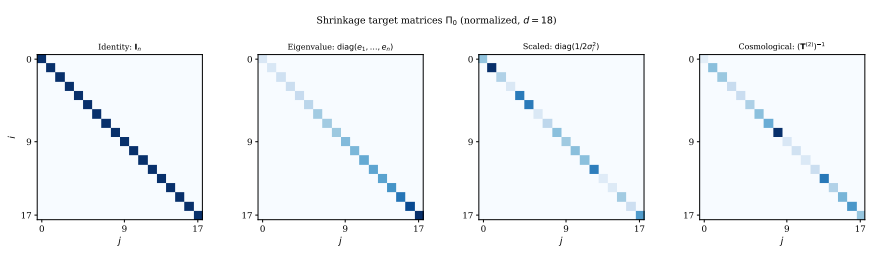
- Embed the *direct precision shrinkage* framework (Bodnar et al.) inside the dual formulation of the Ensemble Kalman Filter (EnKF), recomputing coefficients each cycle.
- Implement and compare **six shrinkage estimators** plus two EnKF baselines on systems with opposite precision structure.
- Propose a **scaled-identity target** that needs no domain-specific information.
- Release all implementations in the open-source TEDA framework.

Methodology

The assimilation cycle. At each step, an ensemble of states is propagated by the model, mixed with new observations, and updated. The covariance of the ensemble determines how observations are weighted.



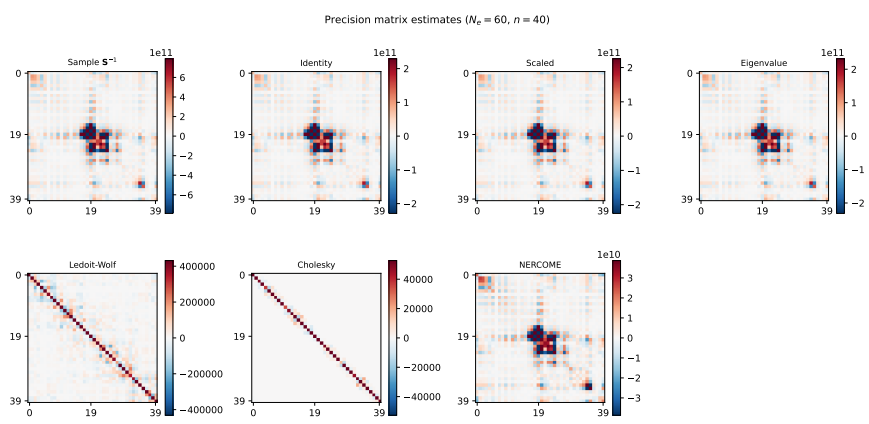
Estimators evaluated. Four *direct precision* targets (identity, scaled-identity *[ours]*, eigenvalue, cosmological), one *covariance-space* method (Ledoit–Wolf), and the nonparametric *NERCOME*. Two baselines: standard EnKF and EnKF–Cholesky.



Shrinkage targets compared: structure imposed on the estimate determines how the noisy sample is regularized.

Test systems. Two systems with opposite precision structure to stress-test target-truth alignment:

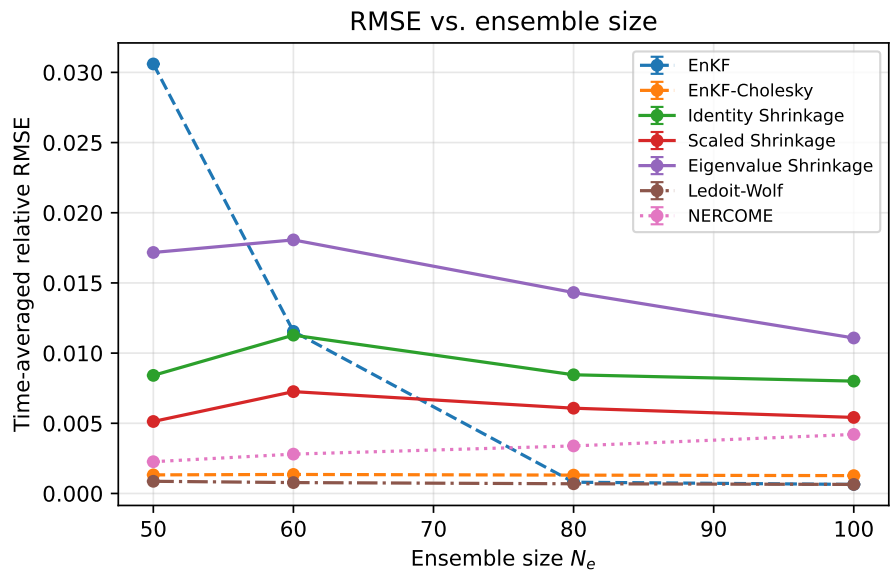
- **Lorenz 96** (n=40): chaotic atmospheric model with *dense* off-diagonal precision from spatial coupling.
- **BOSS DR12 mocks** (d=18): cosmological power-spectrum data with *near-diagonal* precision from independent k-bins.



True precision matrices: Lorenz 96 (left, dense) vs. BOSS DR12 (right, near-diagonal).

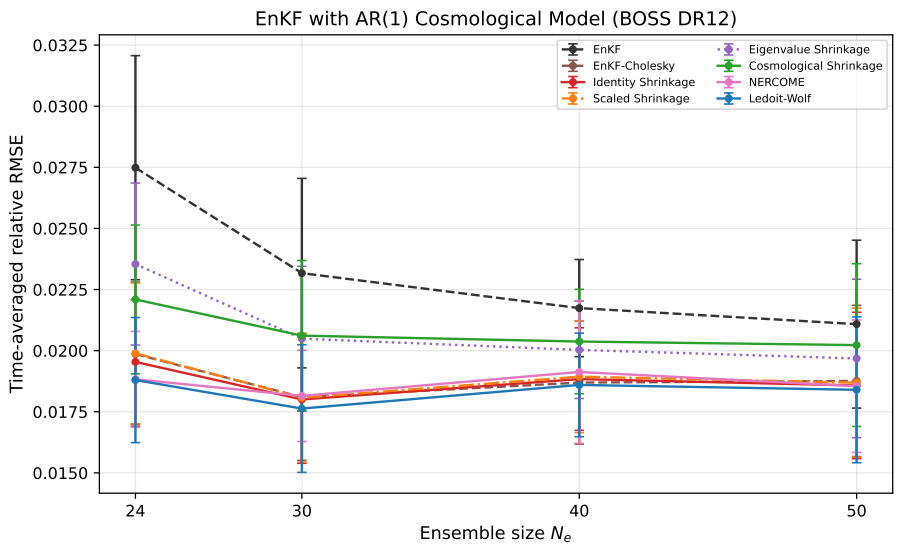
Main Results

Result 1 — target alignment is decisive. Diagonal targets break on Lorenz 96 (dense precision) but excel on cosmological data (near-diagonal precision).



Lorenz 96: RMSE vs. ensemble size. Diagonal-target shrinkage is unstable; Ledoit–Wolf is most stable.

Result 2 — best estimator ≠ best filter. On cosmological data, Ledoit–Wolf produces the *worst* standalone precision estimates yet the *lowest* EnKF RMSE. Low cycle-to-cycle variance matters more than low bias.



Cosmological data: EnKF RMSE across estimators. The most biased standalone estimator wins the sequential filter.

Key Numbers

- **6** shrinkage estimators · **2** EnKF baselines
- Misaligned targets: cond. 10^7–10^{11}, β^* up to 10^9, clamping fires in >**80%** of cycles
- Scaled-identity target: **10–30%** gain over plain identity
- Ledoit–Wolf stdev across cycles: ±0.01 vs. ±0.13 for direct precision

Conclusions

- **Target-truth alignment** drives the success or failure of direct precision shrinkage; there is no one-size-fits-all target.
- **Variance reduction beats bias reduction** for sequential filtering: standalone metrics like Frobenius loss can mislead practitioners choosing an estimator for the EnKF.
- The proposed **scaled-identity target** gives a 10–30% improvement over plain identity on both systems with no domain-specific tuning.
- Practical impact: more reliable forecasts and cosmological parameter estimates from the same limited ensembles.

Future Work

- Combine shrinkage with **spatial localization** for high-dimensional models (weather, $n \sim 10^6$).
- **Adaptive targets** that track the evolving precision structure across cycles.
- Theoretical treatment of the **variance–bias trade-off** in sequential filtering (discrete Lyapunov framework).
- Extension to **nonlinear, non-Gaussian** ensembles.

Resources

Full thesis: `thesis/main.pdf`
Scientific paper (LNCS): `paper/main.pdf`
TEDA framework: `https://github.com/enino84/TEDA`